

Median

①

Median is the middle point or the central value of the variable in a set of observations when observations are arranged either in ascending or in descending order. It is the value of the variable which divides it into two equal parts.

Calculation of Median

1) Ungrouped

Consider $x_1, x_2, \dots, x_n$ are N variable of the observation.

(i) If N is odd then

$$\text{Median} = \left(\frac{N+1}{2} \right)^{\text{th}} \text{ place value}$$

(ii) If N is even then

$$\text{Median} = \frac{\left(\frac{N}{2} \right)^{\text{th}} \text{ place value} + \left(\frac{N}{2} + 1 \right)^{\text{th}} \text{ place value}}{2}$$

Example

1) Find the median of the following numbers:-

(i) 21, 12, 49, 37, 48, 88, 46, 55, 74, 63

(ii) 88, 72, 33, 29, 70, 86, 54, 91, 61

2) The number of runs scored by 11 players of cricket team of R.K. Mission Vidyamandira:

5, 19, 42, 11, 50, 30, 21, 0, 52, 36, 27

Continuous frequency Distribution

In the case of continuous frequency distribution, the class corresponding to the cf. just greater than $\frac{1}{2}N$ is called the median class and the value of median is obtained by the following formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - C.f}{f} \right) \times h$$

where,

- L is the lower limit of the median class.
- f is the frequency of the median class.
- h is the magnitude of the median class.

~~Cf is the c.f. of the class~~
 Cf is the cumulative frequency of preceding class.

Example

Class boundaries	15-25	25-35	35-45	45-55	55-65
Frequency	4	11	19	14	0

Ans = 40.263

65-75
2

2) Find The following table gives the marks obtained by students in Economics. Find the median:

Marks	11-20	21-30	31-40	41-50	51-60	61-70	71-80
n. of students	42	38	125	84	45	36	30

3) The weekly expenditure of 100 families are given below. Find the median of weekly expenditure

Expenditure	0-10	10-20	20-30	30-40	40-50
Frequency	14	23	27	21	15

24.81

③ Compute the median wage from the following persons weekly wages.

75, 205, 315, 34, 340, 1025, 521, 791, 695, 234

② Grouped Data:-

Consider $x_1, x_2, \dots, x_n$ with their frequency $f_1, f_2, f_3, \dots, f_n$ are N variables

(i) Find $\frac{1}{2}N$, where $N = \sum f_i$

(ii) See the cumulative frequency (cf) just greater than $\frac{N}{2}$

(iii) The corresponding value of x is Median.

x	f	cf
x_1	f_1	f_1
x_2	f_2	$f_1 + f_2$
x_3	f_3	$f_1 + f_2 + f_3$
$\vdots$	$\vdots$	$\vdots$
x_n	f_n	$f_1 + f_2 + \dots + f_n$

Question

① Calculate the median for the following data

No. of student	6	16	7	4	2	8
Marks	20	25	50	9	80	40

② Obtain the median for the following frequency distribution:

x :	1	2	3	4	5	6	7	8	9
f :	8	10	11	16	20	25	15	9	6

④ An incomplete Frequency distribution is given below ④

Variable	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	12	30	?	65	?	25	18

Given that median value is 46. Determine the missing frequencies using median formula. when sum of frequency is 229.

45, 46

⑤ An incomplete